ACTIVITY CO4
(FUNCTIONS IN SQL USING PYTHON AND JOIN METHODS)

```
import sqlite3

# Create database
conn = sqlite3.connect('shop.db')
cursor = conn.cursor()

# Create Customers table
cursor.execute("""
CREATE TABLE Customers (
    cust_id INTEGER PRIMARY KEY,
    name TEXT,
    city TEXT
)
""")

# Create Orders table
cursor.execute("""
CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY,
    cust_id INTEGER,
    amount INTEGER,
    FOREIGN KEY(cust_id) REFERENCES Customers(cust_id)
)
""")

# Insert data into Customers table
customers = [
    (1, 'Ravi', 'Pune'),
    (2, 'Priya', 'Delhi'),
    (3, 'Amit', 'Mumbai'),
    (4, 'Kiran', 'Chennai')
]
```

```
cursor.executemany(
    "INSERT INTO Customers VALUES (?, ?, ?)",
    customers
)

# Insert data into Orders table
orders = [
    (101, 1, 2500),
    (102, 2, 1500),
    (103, 1, 3200),
    (104, 5, 4000)
]

cursor.executemany(
    "INSERT INTO Orders VALUES (?, ?, ?)",
    orders
)

conn.commit()

print("Tables created successfully.")
```

```
Tables created successfully.
```

FUNCTIONS IN SQL

1. COUNT()

```
cursor.execute("SELECT COUNT(*) FROM Orders")
result = cursor.fetchone()
print("Total orders:", result[0])
```

```
Total orders: 4
```

2. SUM()

```
cursor.execute("SELECT SUM(amount) FROM Orders")
result = cursor.fetchone()
print("Total amount:", result[0])
```

```
Total amount: 11200
```

3. AVG()

```
cursor.execute("SELECT AVG(amount) FROM Orders")
result = cursor.fetchone()
print("Average amount:", result[0])
```

```
Average amount: 2800.0
```

4. MIN() and MAX()

```
cursor.execute("SELECT MIN(amount), MAX(amount) FROM Orders")
result = cursor.fetchone()
print("Minimum amount:", result[0])
print("Maximum amount:", result[1])
```

```
Minimum amount: 1500
Maximum amount: 4000
```

5. GROUP BY

```
cursor.execute("""
SELECT cust_id, SUM(amount)
FROM Orders
GROUP BY cust_id
""")

rows = cursor.fetchall()

print("\nGROUP BY")

for row in rows:
    print(row)
```

```
GROUP BY
(1, 5700)
(2, 1500)
(5, 4000)
```

6. ORDER BY

```
cursor.execute("""
SELECT * FROM Orders
ORDER BY amount DESC
""")

rows = cursor.fetchall()

print("\nORDER BY")

for row in rows:
    print(row)
```

```
ORDER BY
(104, 5, 4000)
(103, 1, 3200)
(101, 1, 2500)
(102, 2, 1500)
```

JOIN METHODS

7. INNER JOIN

```
cursor.execute("""
SELECT c.name, o.order_id, o.amount
FROM Customers c
INNER JOIN Orders o
ON c.cust_id = o.cust_id
""")

rows = cursor.fetchall()

print("\nINNER JOIN")

for row in rows:
    print(row)
```

```
INNER JOIN
('Ravi', 101, 2500)
('Priya', 102, 1500)
('Ravi', 103, 3200)
```

8. LEFT JOIN

```
cursor.execute("""
SELECT c.name, o.order_id, o.amount
FROM Customers c
LEFT JOIN Orders o
ON c.cust_id = o.cust_id
""")

rows = cursor.fetchall()

print("\nLEFT JOIN")

for row in rows:
    print(row)
```

```
LEFT JOIN
('Ravi', 101, 2500)
('Ravi', 103, 3200)
('Priya', 102, 1500)
('Amit', None, None)
('Kiran', None, None)
```

9. RIGHT JOIN

```
cursor.execute("""
SELECT c.name, o.order_id, o.amount
FROM Orders o
LEFT JOIN Customers c
ON c.cust_id = o.cust_id
""")

rows = cursor.fetchall()

print("\nRIGHT JOIN")

for row in rows:
    print(row)
```

```
RIGHT JOIN
('Ravi', 101, 2500)
('Priya', 102, 1500)
('Ravi', 103, 3200)
(None, 104, 4000)
```

10. FULL JOIN

```
SELECT c.name, o.order_id, o.amount
FROM Customers c
LEFT JOIN Orders o
ON c.cust_id = o.cust_id

UNION

SELECT c.name, o.order_id, o.amount
FROM Orders o
LEFT JOIN Customers c
ON c.cust_id = o.cust_id

""")

rows = cursor.fetchall()

print("\nFULL JOIN")

for row in rows:
    print(row)
```

```
FULL JOIN
(None, 104, 4000)
('Amit', None, None)
('Kiran', None, None)
('Priya', 102, 1500)
('Ravi', 101, 2500)
('Ravi', 103, 3200)
```